Data Scientist (DS)

Module - 4

Assignment 07: Classification Algorithms Model Building

Project Title: Crowdfunding Campaign Success Prediction

Problem Statement

Crowdfunding platforms host thousands of campaigns across categories such as technology, education, healthcare, and creative arts. However, not all campaigns achieve their funding goals. Businesses and investors want to understand which campaign factors influence success and which machine learning model can best predict campaign outcomes.

In this project, you will build and compare multiple classification models to predict whether a crowdfunding campaign will be **Successful** or **Unsuccessful** based on campaign-related features such as funding goal, campaign duration, backer engagement, and pledged amount.

The project focuses on model evaluation, cross-validation, hyperparameter tuning, and selecting the best-performing classification model.

Dataset Link: Crowdfunding Campaign Dataset

Assignment Tasks

Task 1 – Data Exploration & Preprocessing

- Load and inspect the dataset
- Handle missing values and duplicates
- Encode categorical variables
- Perform feature selection
- Split the dataset into training and testing sets
- Apply feature scaling where required

Task 2 – Classification Model Building

Build and train the following classification models:

- Random Forest Classifier
- KNN
- Decision Tree
- Naive Bayes Classifier
- Support Vector Machine (SVM)

Experiment with different model parameters to improve performance.

Task 3 – Model Evaluation

Evaluate all models using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Compare training and testing performance.

Task 4 – Cross-Validation & Hyperparameter Tuning

- Apply Cross-Validation to evaluate model stability
- Perform Hyperparameter Tuning using GridSearchCV or RandomizedSearchCV
- Identify the best parameters for each model

Task 5 – Model Comparison & Selection

- Compare all classification models using evaluation metrics
- Identify the best-performing model
- Discuss strengths, weaknesses, and possible improvements

Deliverables

- Google Colab / Jupyter Notebook with complete code and outputs
- Trained classification models
- Evaluation metrics and comparison results
- Final conclusion identifying the best model
- Google Drive / GitHub submission link with View access

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Level
Data Preprocessing	Data Preparation	Cleaning, encoding, scaling	Intermediate
Model Building	Machine Learning	RF, Naive Bayes, SVM	Intermediate
Model Evaluation	Performance Analysis	Accuracy, Precision, Recall, F1	Intermediate
Cross-Validation	Model Validation	K-Fold validation techniques	Intermediate
Hyperparameter Tuning	Optimization	GridSearchCV / RandomizedSearchCV	Intermediate
Model Comparison	Decision Making	Best model selection	Intermediate

Rubric & Mark Allocation (Total 10 Marks)

Skill / Component	Marks	Full Marks Criteria	Partial Credit
Data Preprocessing	2	Dataset cleaned and prepared correctly	Minor preprocessing issues
Model Building	3	All three models implemented correctly	Missing/incomplete models
Model Evaluation	2	Metrics and confusion matrix generated correctly	Partial evaluation
Cross-Validation & Tuning	2	Proper validation and tuning performed	Incomplete tuning
Model Comparison & Insights	1	Best model identified with proper reasoning	Weak comparison

Skill Taxonomy Structure

- **Data Preparation:** Cleaning, encoding, scaling, train-test split
- **Classification Modeling:** RF, Naive Bayes, SVM implementation
- **Model Evaluation:** Accuracy, Precision, Recall, F1-score
- **Model Validation:** Cross-validation techniques
- **Optimization:** Hyperparameter tuning
- **Reporting:** Compare models and interpret results

Evaluation Guidelines

- Review notebook/code, outputs, and explanations
- Verify preprocessing, training, evaluation, and tuning steps
- Check correctness of metrics and model comparison
- Assign marks based on rubric
- Focus on correctness, clarity, and implementation quality

Achievement Criteria & Passing Rules

- All sections must be attempted
- Students must fully achieve at least two key skills:
    - Model Building
    - Model Evaluation
    - Hyperparameter Tuning
- Models must execute without errors
- Cross-validation and tuning are mandatory
- A detailed scorecard and feedback must be provided for every submission